Accelerated Math II/III Name________________________________

Vertex and Intercept Forms & Characteristics

Date________________ Period____

Identify the vertex and axis of symmetry of each.

1) $y = -2x^2 + 2$

2) $y = -\frac{1}{4}(x - 8)^2 + 8$

3) $y = \frac{1}{2}(x - 9)(x - 7)$

4) $y = -\frac{1}{18}(x - 8)(x - 6)$

5) $y = 4(x - 9)(x + 4)$

6) $y = -2(x - 5)(x + 3)$

Identify the vertex, axis of symmetry, x-intercepts, and y-intercept of each. Then sketch the graph.

7) $f(x) = \frac{1}{3}(x - 5)(x + 1)$

8) $f(x) = \frac{1}{2}(x - 6)(x - 3)$

9) $f(x) = (x + 5)(x + 1)$

10) $f(x) = 2(x - 5)^2$

Identify the vertex, axis of symmetry, and y-intercept of each. Then sketch the graph. Determine the domain, range, and interval of increase and decrease.

11) $y = -x^2 + 8x - 21$

12) $y = (x - 5)^2 - 2$

13) $y = -2(x - 5)^2$

14) $y = 2x(x + 2)$

Answers to Vertex and Intercept Forms & Characteristics

1) Vertex: $(0, 2)$
Axis of Sym.: $x = 0$

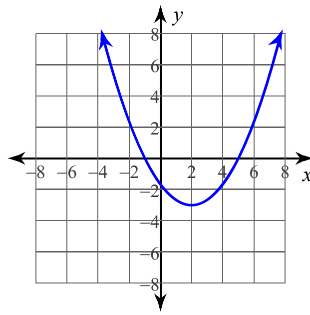
2) Vertex: $(8, 8)$
Axis of Sym.: $x = 8$

3) Vertex: $\left(8, -\frac{1}{2}\right)$
Axis of Sym.: $x = 8$

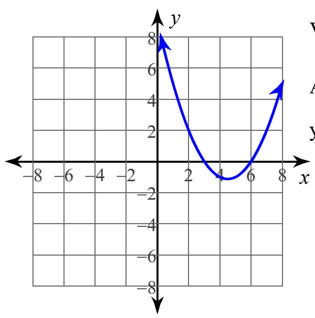
4) Vertex: $\left(7, \frac{1}{18}\right)$
Axis of Sym.: $x = 7$

5) Vertex: $\left(\frac{5}{2}, -169\right)$
Axis of Sym.: $x = \frac{5}{2}$

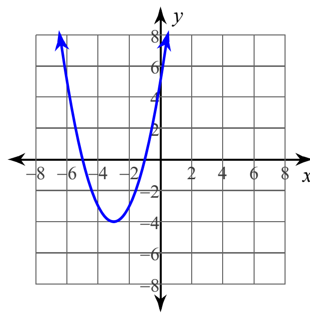
6) Vertex: $(1, 32)$
Axis of Sym.: $x = 1$

7)


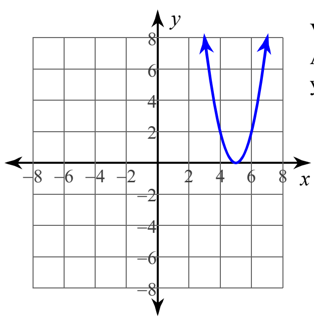
Vertex: $(2, -3)$
Axis of Sym.: $x = 2$
y-int: $-\frac{5}{3}$

8)


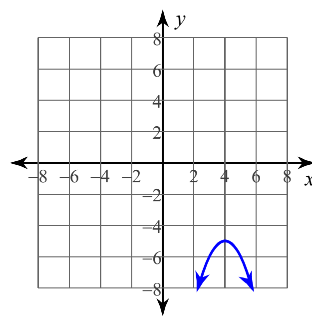
Vertex: $\left(\frac{9}{2}, -\frac{9}{8}\right)$
Axis of Sym.: $x = \frac{9}{2}$
y-int: 9

9)


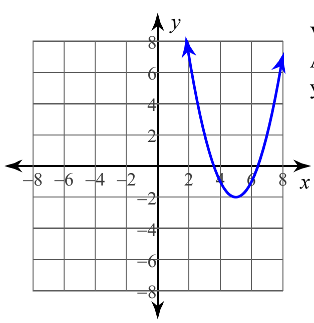
Vertex: $(-3, -4)$
Axis of Sym.: $x = -3$
y-int: 5

10)


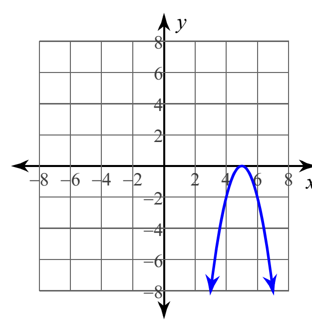
Vertex: $(5, 0)$
Axis of Sym.: $x = 5$
y-int: 50

11)


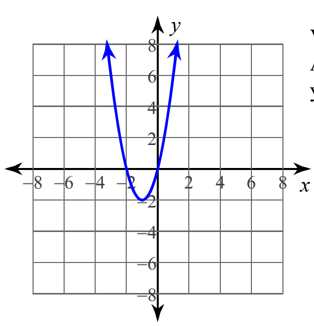
Vertex: $(4, -5)$
Axis of Sym.: $x = 4$
y-int: -21

12)


Vertex: $(5, -2)$
Axis of Sym.: $x = 5$
y-int: 23

13)


Vertex: $(5, 0)$
Axis of Sym.: $x = 5$
y-int: -50

14)


Vertex: $(-1, -2)$
Axis of Sym.: $x = -1$
y-int: 0